

Exploratory Data Analysis Observations for `SLURRY_FREE_ACID` Prediction

TSP Fertilizer Process Analytics Project

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1 Purpose of This Report

This report summarizes the main findings from the exploratory analysis of the one-minute TSP fertilizer production dataset. The confirmed prediction target is:

`SLURRY_FREE_ACID`

The objective is not only to describe the dataset statistically, but also to interpret what the observed patterns imply for future feature engineering, machine learning, and industrial understanding of the process.

The analysis uses only `tsp_1min.csv`. Any other CSV file present in the repository is ignored.

2 Dataset Structure and Time Coverage

2.1 Main Facts

- Number of rows: 44,640.
- Number of columns: 23.
- Time range: from 2026-01-01 00:00 UTC to 2026-01-31 23:59 UTC.
- Sampling frequency: one row per minute.
- Duplicate rows: 0.
- Duplicate timestamps: 0.
- Confirmed target: `SLURRY_FREE_ACID`.

2.2 Interpretation

The dataset covers exactly one full month of minute-level operation. This is a strong foundation for exploratory time-series analysis because the timestamp sequence is clean and regular. The absence of duplicate timestamps is particularly important: it means each row can be interpreted as one unique operating state.

However, one month is still a relatively short historical window for industrial machine learning. It is enough to study short-term behavior, daily patterns, operational disturbances, and preliminary lag relationships. It is not enough to make strong conclusions about yearly seasonality, long-term feedstock changes, equipment aging, or rare events.

2.3 Consequence for Future Work

- Future model validation must respect chronological order.
- Random train-test splitting should not be used because it would mix future plant behavior into the training set.
- A suitable first validation strategy is to train on the earlier part of January and test on the latest part of January.
- If more months become available, the next step should be rolling-origin or expanding-window validation.

3 Missing Values and Completeness

3.1 Observed Pattern

Missing values are present but low overall. The highest missing percentages are below 1%.

Variable	Missing Count	Approx. Missing Percentage
DEPRESSURE_SECHEUR	444	0.995%
FLOW_RATE_FIOUL	414	0.927%
FLOW_RATE_VAPEUR	396	0.887%
FLOW_RATE_PHOSPHORIC_ACID_2	393	0.880%
FLOW_RATE_SLURRY_M3_HR	385	0.862%
RECYCLAGE	368	0.824%
FLOW_RATE_PHOSPHORIC_ACID_1	367	0.822%
SLURRY_FREE_ACID	308	0.690%

3.2 Interpretation

The dataset is not heavily incomplete. Missingness below 1% usually does not prevent modeling, but it still matters because process data is time-dependent. A missing value may represent a communication issue, sensor unavailability, historian drop, or a plant event. Therefore, missing values should not be treated as random without checking their time distribution.

3.3 Consequence for Future Work

- Do not drop all rows with missing values by default; this could remove important event periods.
- Use train-only imputation in machine learning pipelines.
- Median imputation is a reasonable baseline for tree models.
- For time-series models, forward fill or interpolation may be considered, but only if it is physically defensible for the sensor.
- Add missingness indicators for critical sensors if missing values cluster around disturbances.

4 Target Variable: SLURRY_FREE_ACID

4.1 Statistical Summary

Metric	Value
Valid observations	44,332
Missing observations	308
Mean	11.298
Median	11.311
Standard deviation	0.515
Minimum	3.392
1st percentile	10.225
5th percentile	10.536
25th percentile	10.992
75th percentile	11.620
95th percentile	12.012
99th percentile	12.308
Maximum	52.671
Skewness	12.084
Kurtosis	959.040
IQR lower bound	10.050
IQR upper bound	12.562
IQR outliers	352

4.2 Main Pattern

Most SLURRY_FREE_ACID values are concentrated around 11.0 to 11.6. The median is 11.311 and the mean is 11.298, which indicates that the normal operating center is very stable.

At the same time, the maximum value is 52.671, which is extremely far from the main operating range. This single value, along with other spikes such as 25.964 and 20.771, creates very high skewness and kurtosis.

4.3 Important Extreme Values

Timestamp	SLURRY_FREE_ACID
2026-01-18 14:12 UTC	52.671
2026-01-19 13:35 UTC	25.964
2026-01-02 06:41 UTC	20.771
2026-01-13 12:59 UTC	17.487
2026-01-13 23:32 UTC	17.293
2026-01-23 09:11 UTC	15.744
2026-01-01 00:06 UTC	15.118

Very low target values were also observed, including 3.392 on 2026-01-05 05:50 UTC and 6.023 on 2026-01-18 19:50 UTC.

4.4 Interpretation

The target behaves like a mostly stable quality variable with occasional abnormal events. The typical variation is small, but rare excursions are large. This is common in industrial quality data: normal production may be tightly controlled, while startup, shutdown, maintenance, feed changes, instrumentation errors, or process disturbances create unusual values.

The extreme value 52.671 should be treated carefully. It may be:

- a real process upset,
- a laboratory or analyzer anomaly,
- a data-entry or historian issue,
- a transient during nonstandard operation,
- or a unit/scale inconsistency at that timestamp.

4.5 Consequence for Future Work

- Do not automatically remove target outliers.
- Create an event flag for extreme target values and review them with process engineers.
- Consider robust evaluation metrics such as MAE and median absolute error, not only RMSE.
- Compare model performance on normal operation and abnormal operation separately.
- If the extreme target values are confirmed as sensor or lab errors, handle them in a documented cleaning step.
- If the extreme target values are real process events, preserve them because they are important for industrial reliability.

5 Outlier Behavior

5.1 Observed Pattern

Using the IQR rule, 352 rows are flagged as target outliers. Some outliers are isolated spikes, while others occur in short continuous periods.

Examples of outlier periods include:

Start	End	Count	Min Target	Max Target
2026-01-28 22:18	2026-01-28 22:45	28	8.659	9.996
2026-01-13 19:47	2026-01-13 20:09	23	8.625	9.967
2026-01-05 10:25	2026-01-05 10:47	23	9.434	10.039
2026-01-22 08:53	2026-01-22 09:12	20	8.594	9.818
2026-01-28 20:57	2026-01-28 21:11	15	9.867	10.038

5.2 Interpretation

The outlier periods are not only high spikes. Several outlier windows are low-acid periods below the IQR lower bound. This means future modeling must handle both high-acid excursions and low-acid excursions.

Continuous outlier intervals are more likely to represent real process regimes or disturbances than single-point spikes. Single-point extremes may still be real, but they deserve special sensor validation.

5.3 Consequence for Future Work

- Add event-window features: rolling maximum, rolling minimum, rolling standard deviation, and recent change rate.
- Consider separate error analysis for low-acid and high-acid abnormal periods.
- Ask process engineers whether the main outlier windows correspond to known events.
- Avoid training a model that only performs well around the median target value.

6 Negative and Physically Suspicious Values

6.1 Observed Pattern

Several variables contain negative values. Some may be physically impossible depending on sensor definition.

Variable	Negative Count
DEPRESSURE_SECHEUR	4,861
FLOW_RATE_VAPEUR	475
FLOW_RATE_SLURRY_M3_HR_2	453
FLOW_RATE_FIOUL	451
FLOW_RATE_SLURRY_M3_HR	420
FLOW_RATE_LIQUIDE_LAVAGE	360
PRODUCTION_TSP_BALANCE	354
RECYCLAGE	314
FLOW_RATE_GROUND_PHOSPHATE	252
FLOW_RATE_SLURRY_T_PER_HR	89
FLOW_RATE_PHOSPHORIC_ACID_2	59
FLOW_RATE_PHOSPHORIC_ACID_1	1

6.2 Interpretation

Negative values in flow rates and production balance are suspicious because these quantities are usually non-negative in physical operation. However, negative dryer depression may be meaningful depending on the sign convention. For example, `DEPRESSURE_SECHEUR` may represent draft or vacuum, where negative values can be physically valid.

Negative flow values may indicate:

- sensor noise around zero,
- calibration offset,

- shutdown or stopped-flow conditions,
- reverse signal convention,
- historian preprocessing artifacts,
- or abnormal process conditions.

6.3 Consequence for Future Work

- Define physical validity rules variable by variable with process engineers.
- Do not apply one universal negative-value rule to all sensors.
- For true flow rates, consider clipping small negative noise to zero only if confirmed.
- Add binary flags for suspicious negative values before deciding whether to impute or clip.
- Treat negative production or flow values as potential shutdown/transient indicators.

7 Correlation with SLURRY_FREE_ACID

7.1 Strongest Pearson Correlations

The strongest absolute correlations with SLURRY_FREE_ACID are moderate, not extremely high.

Variable	Pearson Correlation with Target
FLOW_RATE_FIOUL	0.364
FLOW_RATE_VAPEUR	0.359
RECYCLAGE	0.356
FLOW_RATE_SLURRY_T_PER_HR	0.346
FLOW_RATE_SLURRY_M3_HR	0.342
PRODUCTION_TSP_BALANCE	0.339
PRESSURE_VAPEUR	0.323
FLOW_RATE_GROUND_PHOSPHATE	0.319
TEMPERATURE_GAZ_SORTIE_SECHEUR	0.314
LEVEL_CUVE_PASSAGE	0.301
TEMPERATURE_AIR_CHAUD	0.295
FLOW_RATE_PHOSPHORIC_ACID_2	0.289

7.2 Interpretation

All major correlations with the target are positive in this dataset. This suggests that higher production intensity, higher recycle, higher steam/fuel usage, and higher slurry flow tend to be associated with higher free acid.

However, the correlations are moderate. This is important: SLURRY_FREE_ACID is not explained by one single variable. It is likely controlled by a combination of feed ratios, residence time, thermal state, recycle, and operating regime.

7.3 Process Meaning

- FLOW_RATE_FIOUL and thermal variables suggest that dryer or heat-load conditions are linked to acid behavior.
- FLOW_RATE_VAPEUR, PRESSURE_VAPEUR, and steam-related variables suggest utility conditions may influence or track the process state.
- RECYCLAGE and slurry flows suggest that internal circulation and material loading are important.
- Acid flow alone is not the only dominant signal; therefore acid concentration likely depends on balance between acid, phosphate, recycle, and residence time.

7.4 Consequence for Future Work

- Models should include multiple process areas: feed, slurry, recycle, steam, dryer, and production variables.
- Feature engineering should include ratios and interactions, not only raw values.
- Correlation should be used for screening, not for causal claims.
- A tree-based model or gradient boosting model is likely more suitable than a purely linear model as a first strong baseline.

8 Time-Lag Relationships

8.1 Observed Pattern

Several variables show slightly stronger correlation with SLURRY_FREE_ACID when shifted by a few minutes.

Variable	Best Lag (min)	Lagged Corr.	Lag-0 Corr.
FLOW_RATE_VAPEUR	3	0.361	0.359
RECYCLAGE	5	0.360	0.356
FLOW_RATE_FIOUL	1	0.356	0.364
FLOW_RATE_SLURRY_T_PER_HR	1	0.353	0.346
FLOW_RATE_SLURRY_M3_HR	1	0.349	0.342
PRODUCTION_TSP_BALANCE	2	0.340	0.339
PRESSURE_VAPEUR	10	0.327	0.323
FLOW_RATE_GROUND_PHOSPHATE	10	0.324	0.319
TEMPERATURE_GAZ_SORTIE_SECHEUR	2	0.314	0.314
LEVEL_CUVE_PASSAGE	2	0.304	0.301

8.2 Interpretation

The lag improvements are present but small. This means the process has some delayed effects, but the simple lagged correlation method does not reveal a very strong single dead time. The best lags are mostly between 1 and 10 minutes.

This is still meaningful in an industrial process. A one-minute or five-minute lag may represent:

- material residence time,
- analyzer response delay,
- control-loop adjustment,
- transportation delay between process units,
- or the fact that variables are measured at different physical locations.

8.3 Consequence for Future Work

- Include lagged variables at 1, 2, 3, 5, 10, 15, 30, and 60 minutes.
- Include rolling means and rolling standard deviations to capture recent operating history.
- Do not rely only on lag-0 features.
- Use time-aware validation to confirm whether lagged features improve prediction on future data.
- Investigate whether a different target sampling or lab/analyzer delay exists.

9 Rolling Stability

9.1 Observed Pattern

The one-hour rolling standard deviation of `SLURRY_FREE_ACID` has the following approximate quantiles:

Quantile	One-Hour Rolling Std.
50%	0.183
75%	0.213
90%	0.253
95%	0.299
99%	0.817

9.2 Interpretation

Most of the time, target volatility is low. The jump from the 95th percentile to the 99th percentile is large, showing that a small number of periods are much more unstable than normal operation.

This confirms the idea that the process is usually stable, with occasional disturbance windows.

9.3 Consequence for Future Work

- Build features that describe recent volatility.
- Evaluate the model separately during stable and unstable windows.
- Consider an alarm or anomaly layer before prediction if the objective includes operational monitoring.
- During high-variability periods, a model may need different uncertainty bounds.

10 Seasonality and Time-of-Day Patterns

10.1 Observed Pattern

Hourly mean SLURRY_FREE_ACID varies between approximately 11.088 and 11.555.

- Lowest hourly mean: hour 08, approximately 11.088.
- Highest hourly mean: hour 01, approximately 11.555.
- Hourly range: approximately 0.468.

Daily mean target varies more strongly:

- Lowest daily mean: January 18, approximately 10.687.
- Highest daily mean: January 9, approximately 12.026.
- Daily range: approximately 1.339.

10.2 Interpretation

The one-month dataset suggests that daily operating conditions vary more than average hour-of-day effects. Hourly patterns may reflect shifts, routines, sampling practices, or utility cycles, but the dataset is too short to make strong claims.

January 9 and January 18 appear to be important days for process behavior. January 18 is especially notable because it contains both a very high target spike and a very low daily mean.

10.3 Consequence for Future Work

- Add time features such as hour, day of week, and shift if operationally meaningful.
- Do not overinterpret seasonality from one month.
- Investigate plant logs for January 9, January 13, January 18, January 22, and January 28.
- If more history is collected, test weekly and monthly patterns again.

11 Sensor Stability and Health

11.1 Stable Variables

The most stable variables by coefficient of variation include:

Variable	Coefficient of Variation
TEMPERATURE_VAPEUR	0.035
SLURRY_FREE_ACID	0.046
TEMPERATURE_CUVE_ATTACHE	0.106
TEMPERATURE_CUVE_DE_PASSAGE	0.110
TEMPERATURE_SLURRY_PULVERISATEUR	0.110
TEMPERATURE_AIR_CHAUD	0.110
PRESSURE_VAPEUR	0.125
TEMPERATURE_GAZ_SORTIE_SECHEUR	0.125

11.2 Highly Variable Variables

The most variable variables include:

Variable	Coefficient of Variation
DEPRESSURE_SECHEUR	0.869
FLOW_RATE_SLURRY_M3_HR_2	0.617
FLOW_RATE_PHOSPHORIC_ACID_1	0.445
FLOW_RATE_PHOSPHORIC_ACID_2	0.298
PRODUCTION_TSP_BALANCE	0.217
FLOW_RATE_GROUND_PHOSPHATE	0.217
RECYCLAGE	0.216

11.3 Interpretation

Temperature and pressure variables are relatively stable, while flow-related variables vary more. This makes sense industrially: production load, feed rates, recycle, and fuel can change with operating rate, while many temperatures are controlled around setpoints.

FLOW_RATE_SLURRY_M3_HR_2 has very high relative variation and weak correlation with the target. It should be reviewed to understand whether it is a secondary line, redundant sensor, noisy signal, or different measurement point.

11.4 Consequence for Future Work

- Stable temperature variables may still be useful, but mostly for detecting deviations from normal operation.
- Highly variable flow variables are likely important for predicting operating regime.
- Review redundant or alternate slurry flow sensors before modeling.
- Use robust scaling for methods sensitive to variable scale.

12 Multicollinearity and Redundant Sensors

12.1 Observed Pattern

Many process variables are extremely correlated with each other. Examples with absolute correlation above 0.90 include:

Variable A	Variable B	Abs. Corr.
TEMPERATURE_CUVE_DE_PASSAGE	TEMPERATURE_SLURRY_PULVERISATEUR	0.997
TEMPERATURE_CUVE_ATTAQUE	TEMPERATURE_CUVE_DE_PASSAGE	0.996
FLOW_RATE_SLURRY_T_PER_HR	FLOW_RATE_SLURRY_M3_HR	0.996
TEMPERATURE_CUVE_ATTAQUE	TEMPERATURE_SLURRY_PULVERISATEUR	0.993
PRODUCTION_TSP_BALANCE	FLOW_RATE_GROUND_PHOSPHATE	0.974
FLOW_RATE_GROUND_PHOSPHATE	FLOW_RATE_SLURRY_T_PER_HR	0.970
TEMPERATURE_AIR_CHAUD	TEMPERATURE_BRIQUE	0.969

12.2 Interpretation

This is a strongly coupled industrial process. High correlation among variables may reflect:

- common production load,
- mass balance relationships,
- multiple sensors measuring the same physical zone,
- control loops moving variables together,
- thermal coupling between reactor, dryer, and air temperature,
- or redundant instrumentation.

12.3 Consequence for Future Work

- Linear models may suffer from unstable coefficients due to multicollinearity.
- Tree models are less sensitive to multicollinearity but feature importance may be split among correlated variables.
- Feature selection should combine statistics and process knowledge.
- Consider creating representative variables or ratios from highly correlated groups.
- Do not interpret model coefficients as direct physical effects without handling collinearity.

13 Feature Engineering Ideas

13.1 Most Important Feature Families

Based on the observed patterns, future features should represent:

1. **Mass balance:** acid flow, phosphate flow, slurry flow, production, and recycle.
2. **Thermal state:** steam pressure, steam temperature, fuel flow, hot air temperature, dryer outlet gas temperature, brick temperature.
3. **Residence-time memory:** lagged values at 1, 2, 3, 5, 10, 15, 30, and 60 minutes.
4. **Stability:** trailing rolling mean, standard deviation, minimum, maximum, and range.
5. **Disturbance dynamics:** first differences and percentage changes.
6. **Ratios:** acid-to-phosphate, recycle-to-phosphate, slurry density proxy, fuel-to-temperature proxy.
7. **Operating regime:** cluster labels or event flags if validated by process knowledge.

13.2 Specific Candidate Features

- TOTAL_PHOSPHORIC_ACID_FLOW
- ACID_TO_PHOSPHATE_RATIO
- SLURRY_DENSITY_PROXY_T_PER_M3
- RECYCLE_TO_PHOSPHATE_RATIO
- FUEL_TO_HOT_AIR_TEMP_RATIO
- ATTACK_TO_PASSAGE_TEMP_DELTA
- lagged feed and recycle variables,
- rolling target-free process history variables.

13.3 Leakage Warning

The target `SLURRY_FREE_ACID` must never be used as an input feature when predicting itself at the same timestamp. If lagged target values are considered later, they must be used only in a valid forecasting setup where past target values are available at prediction time.

14 Machine Learning Consequences

14.1 Recommended Modeling Direction

The dataset is suitable for a first predictive modeling notebook, but the modeling should be leakage-safe and time-aware.

Recommended model sequence:

1. Naive baseline using previous target value if available in a forecasting setting.
2. Linear regression, Ridge, or Elastic Net as transparent baselines.
3. Random Forest and Extra Trees for nonlinear baselines.
4. Gradient boosting models such as XGBoost, LightGBM, or CatBoost.

14.2 Recommended Validation

- Use chronological split.
- Use rolling-origin validation when more data is available.
- Evaluate normal and abnormal periods separately.
- Track MAE, RMSE, R^2 , and error quantiles.
- Include prediction intervals or uncertainty estimates if we end up building a model that supports operational decisions.

14.3 Preprocessing Recommendations

- Parse and sort timestamps.
- Impute missing values using only training data.
- Review negative values sensor by sensor.
- Create event flags for abnormal target periods and suspicious sensor behavior.
- if we work with PCA, clustering, linear models or distance-based models, we should scale features.
- No centered rolling windows because they use future information.

15 Industrial Understanding

15.1 What the Data Suggests

The process appears to be usually stable around a free-acid operating center near 11.3. The main prediction challenge is not the normal center of the distribution; it is understanding and predicting deviations from that center.

The strongest relationships are not purely acid-feed variables. Instead, free acid is associated with production load, recycle, steam, fuel, slurry flow, and dryer/thermal state. This suggests that `SLURRY_FREE_ACID` is a system-level quality result rather than a simple one-input response.

15.2 Most Important Questions for Process Engineers

1. What happened on January 18 around the extreme target spike?
2. Are the very high target values real laboratory/analyzer readings or data errors?
3. Are negative flow and production values meaningful shutdown indicators or sensor artifacts?
4. Which variables are manipulated by operators and which are only measured responses?
5. What is the expected residence time between acid/phosphate feed changes and slurry free acid response?
6. Is `FLOW_RATE_SLURRY_M3_HR_2` a redundant sensor, secondary stream, or different measurement point?
7. Are January 9, January 13, January 18, January 22, and January 28 associated with known events?

16 Final Summary

- The dataset is structurally strong: one full month, one-minute sampling, no duplicate rows, and no duplicate timestamps.
- Missing values are low, but should still be handled carefully in time-aware pipelines.
- `SLURRY_FREE_ACID` is usually stable near 11.3 but has rare extreme deviations.
- Outliers are industrially meaningful candidates, not automatic deletion candidates.
- Many variables are moderately correlated with the target; no single variable dominates.

- Lagged effects exist but are modest, mostly around 1 to 10 minutes.
- Strong multicollinearity is present because the plant variables move together under common operating regimes.
- Models should use lag features, rolling features, ratios, interaction terms, and time-aware validation.
- The next technical step is a leakage-safe feature engineering and modeling notebook focused on predicting `SLURRY_FREE_ACID`.